

Gem, the GUI that beat Windows to market, goes open source

And Caldera's move makes a certain writer all misty-eyed...

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Gem, the PC GUI that beat Windows to market, has finally gone open source. It was originally developed in 1984, when Windows was still a shaky beta. Current Gem owner Caldera has now made it available under a GNU public licence. Caldera inherited it from Novell, which in turn obtained it when Novell acquired Digital Research, where it was developed. Following the disbanding of the Andover, UK-based Caldera Thin Client team ([see separate story](#)), Caldera decided the time was right to make Gem open source - Caldera no longer has any particular plans for it. The GEM source code is being made available by [Tim Olmstead at the Finnish site](#), with two US mirror sites. Gem was developed at the instigation the late Gary Kildall, the originator of CP/M. In many ways Kildall was the founder of the open software movement, for he was the first to isolate system-specific hardware interfaces in a set of basic I/O system routines - the bios - so that applications could be machine independent. Kildall also developed the filing system and data structures for CD-ROMs. GEM was released in October 1984, five months after a similar but less-flexible interface from Quarterdeck called DESQ (later DESQview). Apple, inspired by the Xerox PARC work, developed its GUI, and Bill Gates was shown a prototype by Steve Jobs in the summer of 1981. In September, Gates initiated the Interface Manager project, which was to be called Windows from 1983. Along the way, Microsoft benefited from being able to examine in detail a Xerox Star (a commercial derivative of the Alto) which Gates bought for \$100,000 in 1981, and a prototype of the Mac in early 1982. Meanwhile, VisiCorp had developed VisiOn, a GUI that worked with MS-DOS. At the 1982 Comdex show, where it was first seen, Gates was reported to be transfixed and to have watched the demo three times. In 1985, IBM produced TopView, which it had announced in 1983, but it was unsuccessful. One of the hitherto-secret Gem documents available for download contains fascinating references to Gem DOS (version 13) for the Motorola 68000. This requires just 128K RAM, 160K ROM, a video controller card with 320 by 200 resolution, and a mouse. It is dated May 1985. (DR-DOS, for Intel processors of course, first appeared in May 1988, when Microsoft was offering MS-DOS 3.) There is even a German version of Gem, and some Turbo Pascal libraries commented in German. The Gem Desktop version 3 requires up to 1.6 megabytes.

Gem was also modified for the Atari and the Tandy 1000 series. There are Gem applications such as Draw, Graph, Paint and Write. The downloads provide a fascinating insight as to how it might have been, had Digital Research decided to compete with MS-DOS in 1985, but the decision to compete was taken two or three years later by Richard Williams, the new CEO, after Gary Kildall had left. It is likely that Gem will be dusted down and incorporated into software for special situations, where fast rendering, small size, and above all, low cost, are important. It provides a useful alternative to X-Windows - and it's free. ®